

Relational Knowledge Distillation for End-to-End Multimodal Audio-LLM Compression

Prepared by: Abhinandan Jaiswal
RV College of Engineering, Bangalore
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ABSTRACT

Traditional audio-understanding models depend heavily on cascaded architectures, pipelining Automatic Speech Recognition (ASR) outputs into Large Language Models (LLMs). This introduces a severe information bottleneck, stripping crucial acoustic features like emotion, tone, and ambient noise. This project defines a novel end-to-end framework that directly projects continuous acoustic embeddings into a specialized language decoder. To deploy this architecture on edge devices, we implement Relational Knowledge Distillation (RKD) on intermediate feature maps. Unlike traditional Knowledge Distillation, which suffers from cross-modal alignment collapse during downsampling, RKD preserves the structural geometric space of a frozen teacher model, ensuring robust semantic alignment even for complex tasks like code-switched Hinglish reasoning in noisy environments.

1. Introduction & Motivation

The paradigm of multimodal machine learning has shifted heavily toward unified models. However, standard methodologies for processing speech involve transcribing audio to text before reasoning. This cascaded method fails in real-world, ambient edge conditions. Nuances such as speaker emotion, background context (e.g., sirens, machinery), and code-switched vernaculars (mixing Hindi and English organically) are lost the moment the audio is forced through a discrete text tokenizer.

While massive end-to-end multimodal models exist, their multi-billion parameter footprints render them unusable on consumer hardware. Therefore, the core challenge this project addresses is model compression across modalities. Cross-modal token downsampling traditionally induces catastrophic alignment collapse. This report details an architecture relying on geometric invariance to solve this problem.

2. Proposed Architecture

The proposed system eliminates discrete intermediate tokenization, establishing a continuous latent bridge between audio perception and language reasoning.

2.1 Acoustic Front-End and Continuous Projection

The input stage utilizes a self-supervised audio encoder designed to extract dense acoustic frames representing rich, contextual embeddings. Instead of quantization, these hidden frames pass through a parameterized **Projection Layer**. This bridge maps continuous audio dimensionality directly into the input embedding manifold of a compact sub-2B parameter LLM decoder.

2.2 The Compact Decoder

The target language decoder acts as the foundational reasoning engine. By bypassing text tokenizers, the LLM is forced to learn to "hear" the continuous embeddings natively, significantly reducing inference latency.

3. Methodology: Relational Knowledge Distillation (RKD)

To train this compact student model efficiently, it must inherit the capabilities of a massive, frozen Teacher Audio-LLM. Standard point-wise distillation fails due to temporal misalignment between modalities.

The proposed solution is Relational Knowledge Distillation (RKD). RKD extracts intermediate hidden-layer feature maps via custom PyTorch forward hooks. It enforces geometric invariance, ensuring the student replicates the multi-dimensional structure of the teacher's latent space.

$$L_{total} = L_{CE} + \alpha L_{RKD-Distance} + \beta L_{RKD-Angle}$$

- **Distance Penalty** ($L_{RKD-Distance}$): Enforces that the normalized Euclidean distances between token clusters in the student model match those in the teacher model.
- **Angular Penalty** ($L_{RKD-Angle}$): Penalizes discrepancies in the cosine angles formed by triplets of tokens, preserving the complex high-dimensional geometry.

4. Implementation Strategy

The development of this architecture is designed to be executable within a standard research timeline (August to September). The core pipeline leverages PyTorch alongside Hugging Face's PEFT libraries. To accommodate hardware constraints, the student decoder will be optimized utilizing Low-Rank Adaptation (LoRA) with $r = 16$ and NF4 quantization, preventing catastrophic forgetting during cross-modal fine-tuning.

Key Evaluation Benchmarks:

1. *Ambient Robustness*: Validation on zero-shot code-switched (Hinglish) dialogue containing controlled background noise profiles.
2. *Efficiency Metrics*: Empirical tracking of absolute parameter reduction, memory footprint compression (MB), and inference latency improvements.

5. Expected Outcomes

This project expects to yield a fully functional, compressed end-to-end audio reasoning framework capable of outperforming comparably sized cascaded models in latency and contextual accuracy. Furthermore, it will provide mathematical and empirical validation that structural geometric distillation avoids the alignment collapse inherent in cross-modal downsampling.